CASE STUDY

Can AI Predict Heart Disease?

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The Dataset

Thirteen features. One Outcome.

Each patient is described by 13 clinical measurements. We predict a single binary outcome (healthy vs disease).

DEMOGRAPHIC

age	in years
sex	0 = female, 1 = male

RESTING VITALS

trestbps	resting blood pressure (mm Hg)
chol	serum cholesterol (mg/dl)
fbs	fasting blood sugar > 120
restecg	resting ECG result
cp	chest pain type (4 categories)

STRESS TEST

thalach	max heart rate achieved
exang	exercise-induced angina
oldpeak	ST depression on exercise
slope	slope of ST segment
ca	# major vessels (fluoroscopy)
thal	thalassemia indicator

→ TARGET: condition 0 = healthy | 1 = has heart disease

THE METHOD

How we built the model.

01	02	03	04	05	06
Explore	Engineer	Split	Scale	Train	Validate

FINDINGS

What the data told us.

C A S E   S T U D Y

Finding 1 - sex

Men develop disease at twice the rate.

F E M A L E

26%

diagnosed with disease

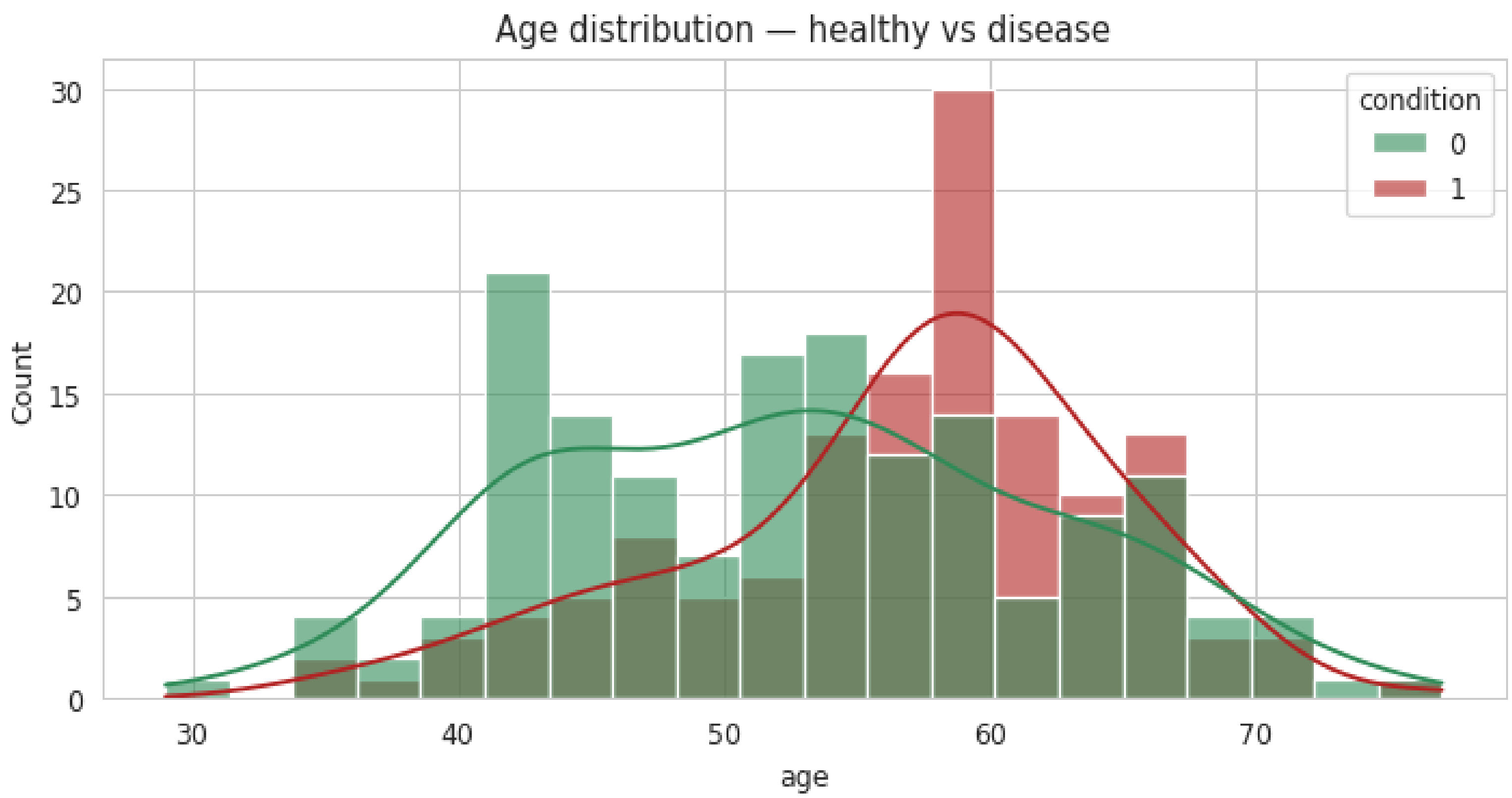
M A L E

55%

diagnosed with disease

Disease shifts older but with massive overlap.

Age is informative but not sufficient. The two distributions overlap heavily.

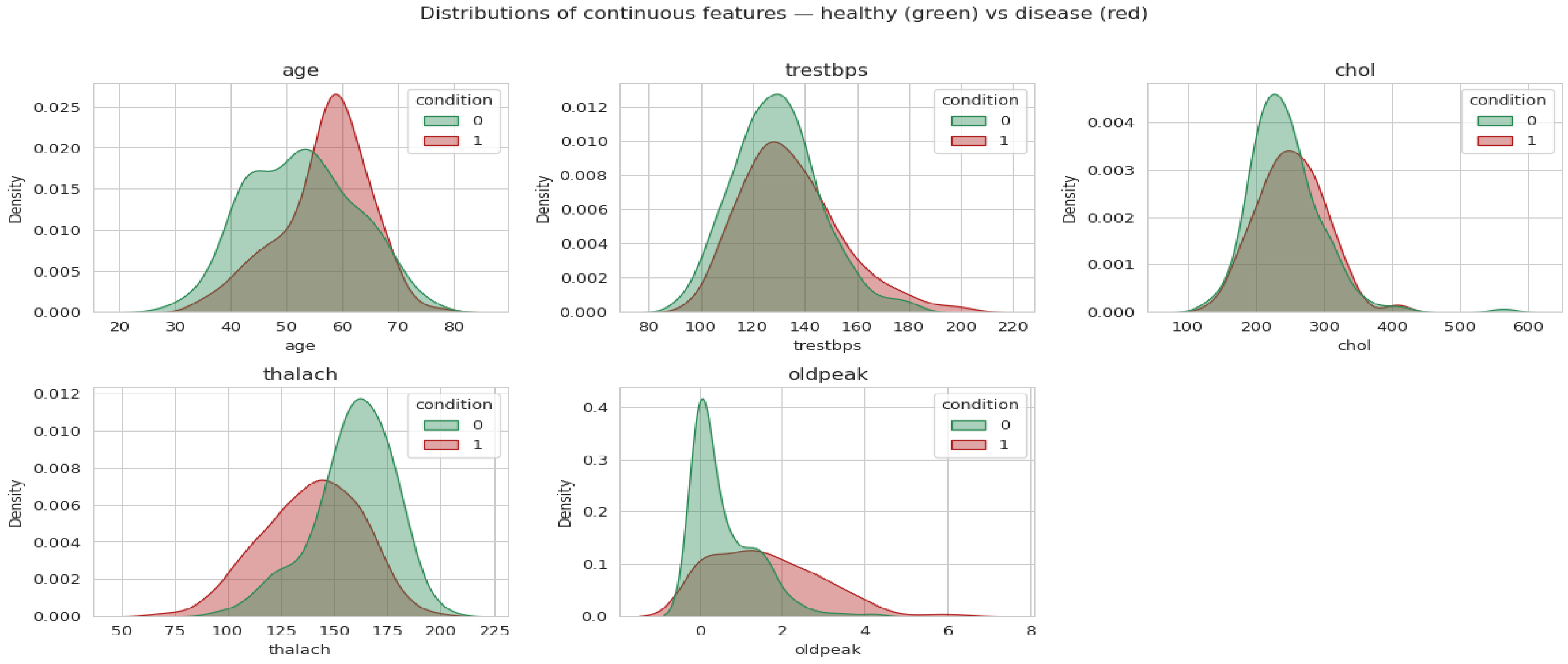


C A S E   S T U D Y

Finding 3 - stress test biomarkers

The stress test separates classes far better than vitals.

Resting BP and cholesterol overlap heavily. But maximum heart rate (thalach) and ST depression (oldpeak) cleanly distinguish patients.



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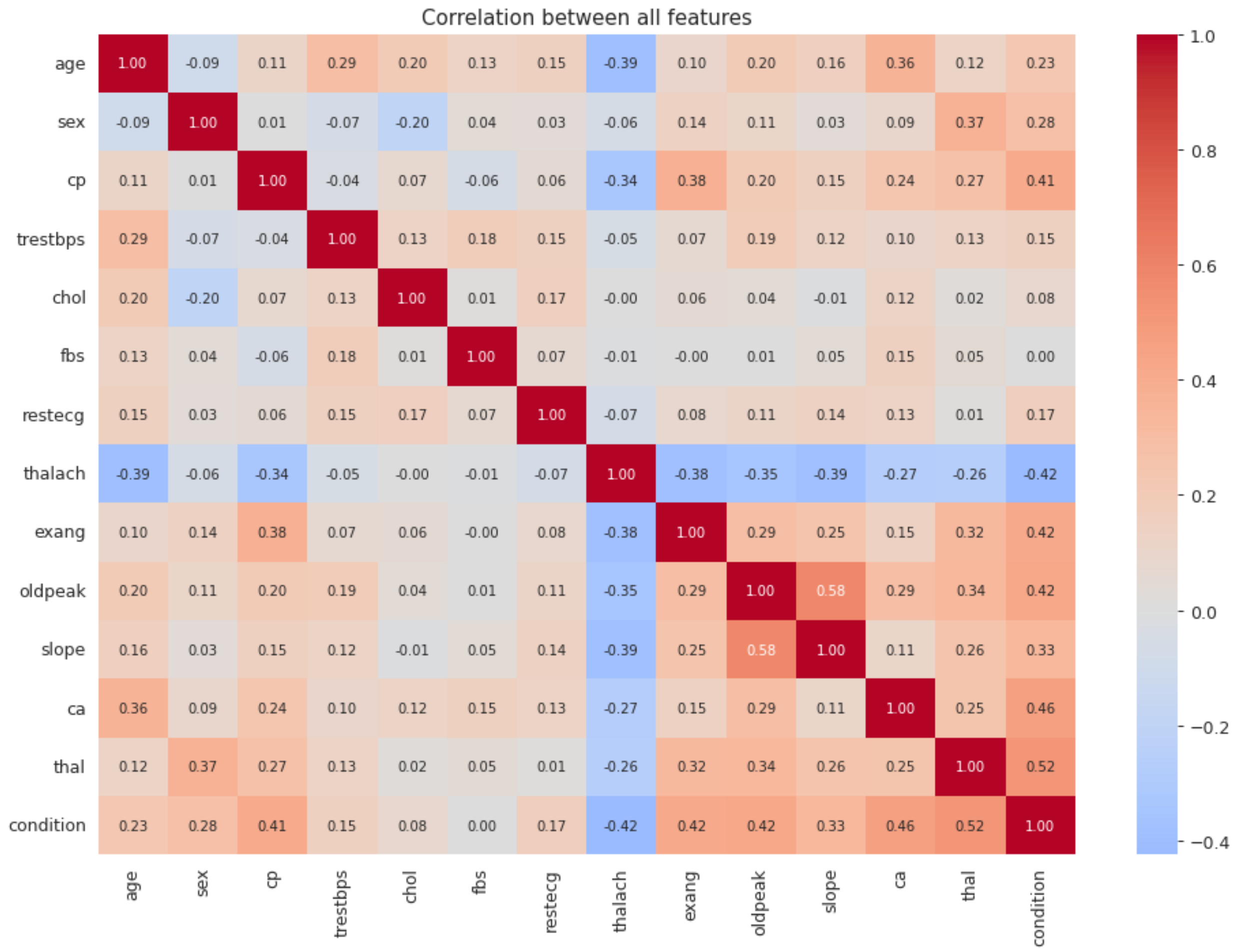
Finding 4 - what correlates with disease

The strongest predictors.



TOP CORRELATIONS WITH "condition"

In order of strength.

thal	+	0.52
ca	+	0.46
exang	+	0.42
oldpeak	+	0.42
thalach	−	0.42
cp	+	0.41
slope	+	0.33

MODEL

Training models.

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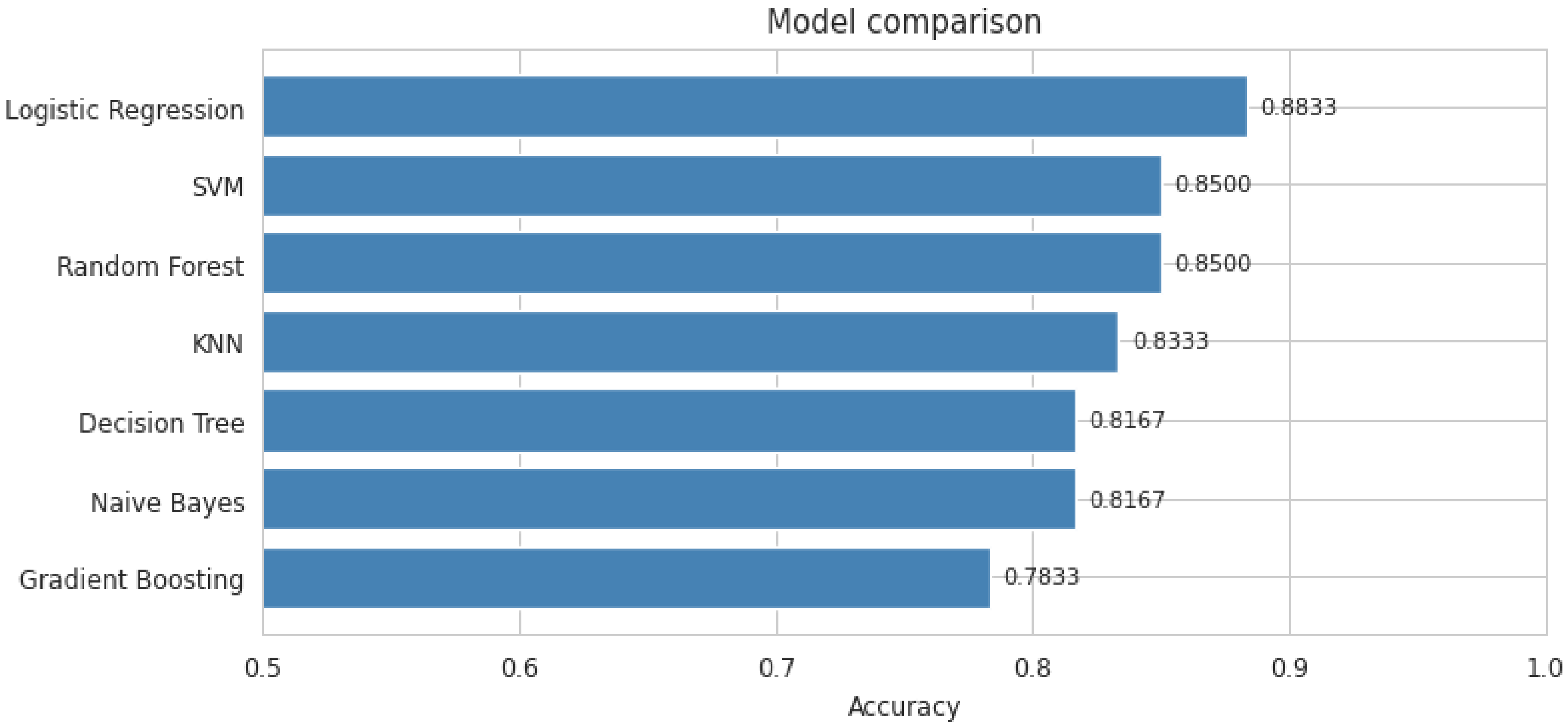
Model Comparison

Logistic Regression wins.

Same train/test split, same features, same scaling only the model changes.



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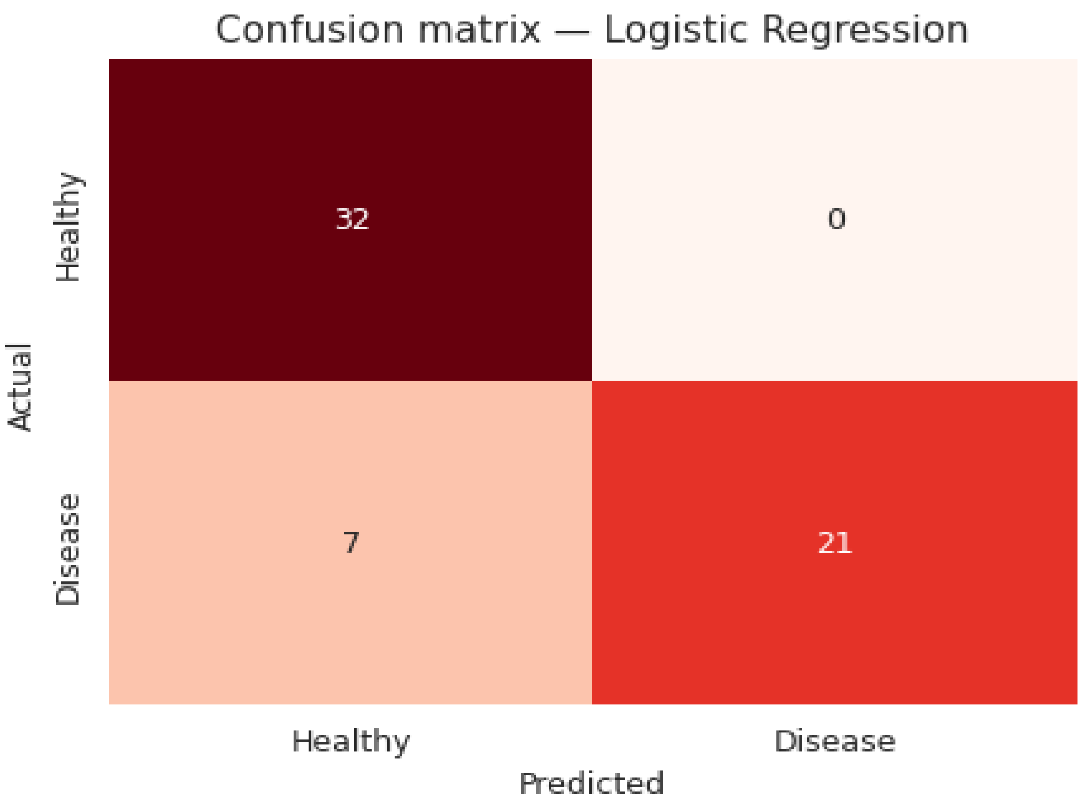
The Evaluation

Where the 88.3% came from.

60 test patients. The model got 53 right. But the 7 wrong ones tell the more important story.



True Negative

32

Correctly healthy

False Positive

0

Wrong alarm

False Negative

7

MISSED DIAGNOSIS

True Positive

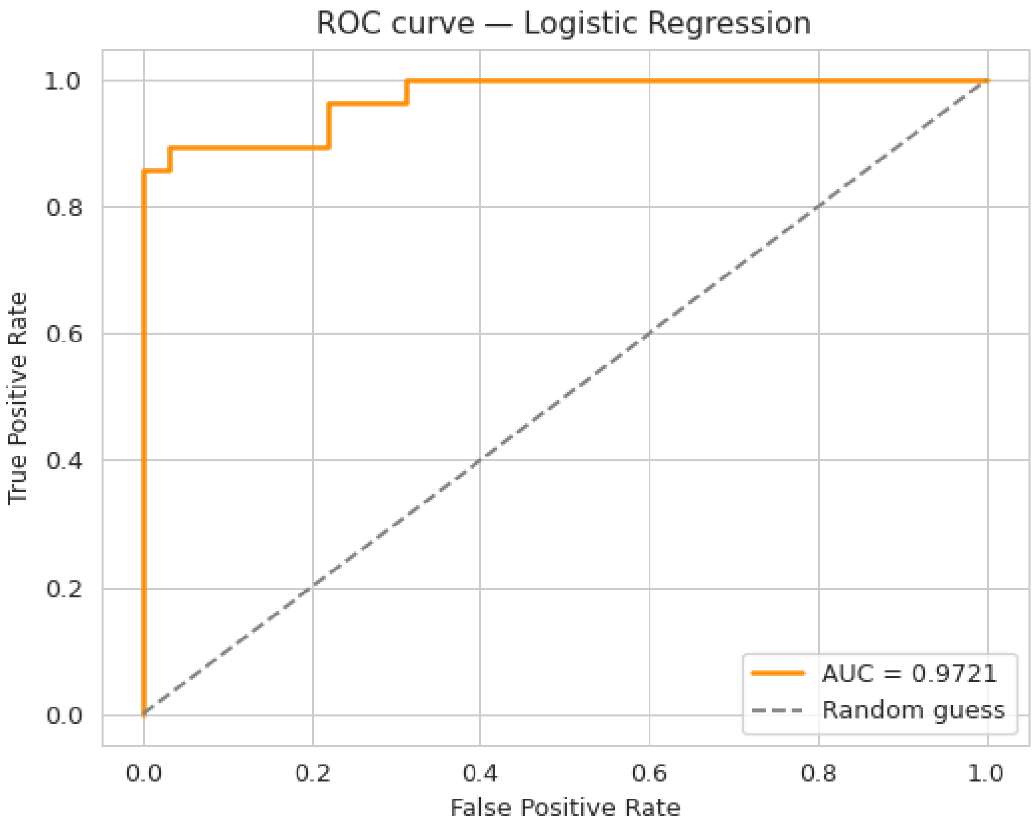
21

Correctly caught disease

In medicine, false negatives are the most dangerous error: a sick patient told they are healthy.

How well can the model rank healthy vs disease?

The ROC curve answers this. The closer the orange curve to the top-left corner, the better.



AREA UNDER CURVE

0.97

1.00	Perfect ranking
0.97	**Our model**
0.80	Good
0.50	Random guess

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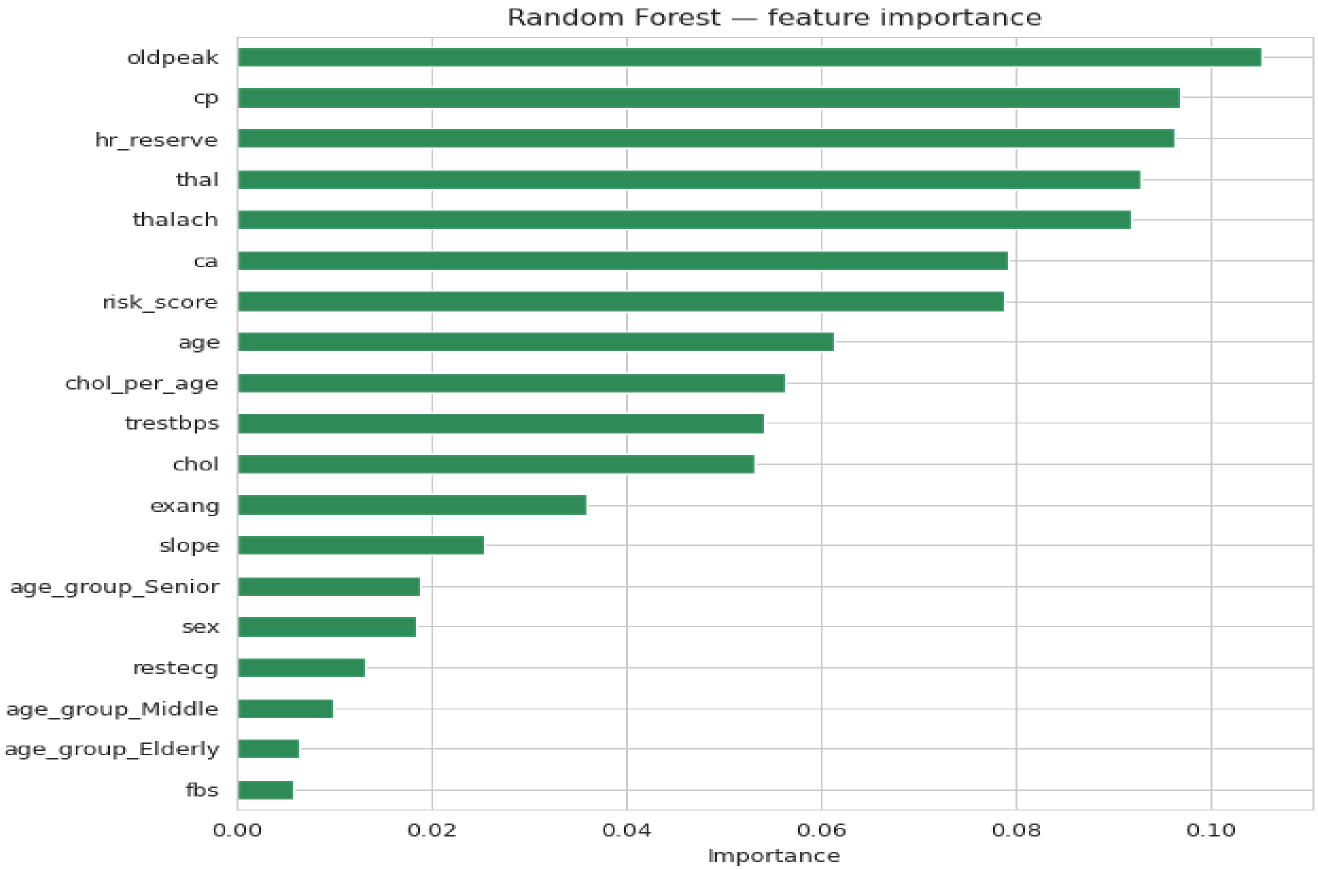
What the model learned

The features that matter most.

Random Forest feature importance, ranked. Our engineered features ended up in the top 7.



THE TAKEAWAYS

Stress-test features dominate

oldpeak, cp, thalach, ca - all stress-test or chest-pain related.

Engineered features made it

hr_reserve and risk_score landed in the top 7. Domain knowledge wins.

Cholesterol is weaker than expected

chol ranks 11th. Surprising - but matches recent cardiology research.

Sex matters less in importance

Even though disease rates differ by sex, clinical signals dominate.

Thank You